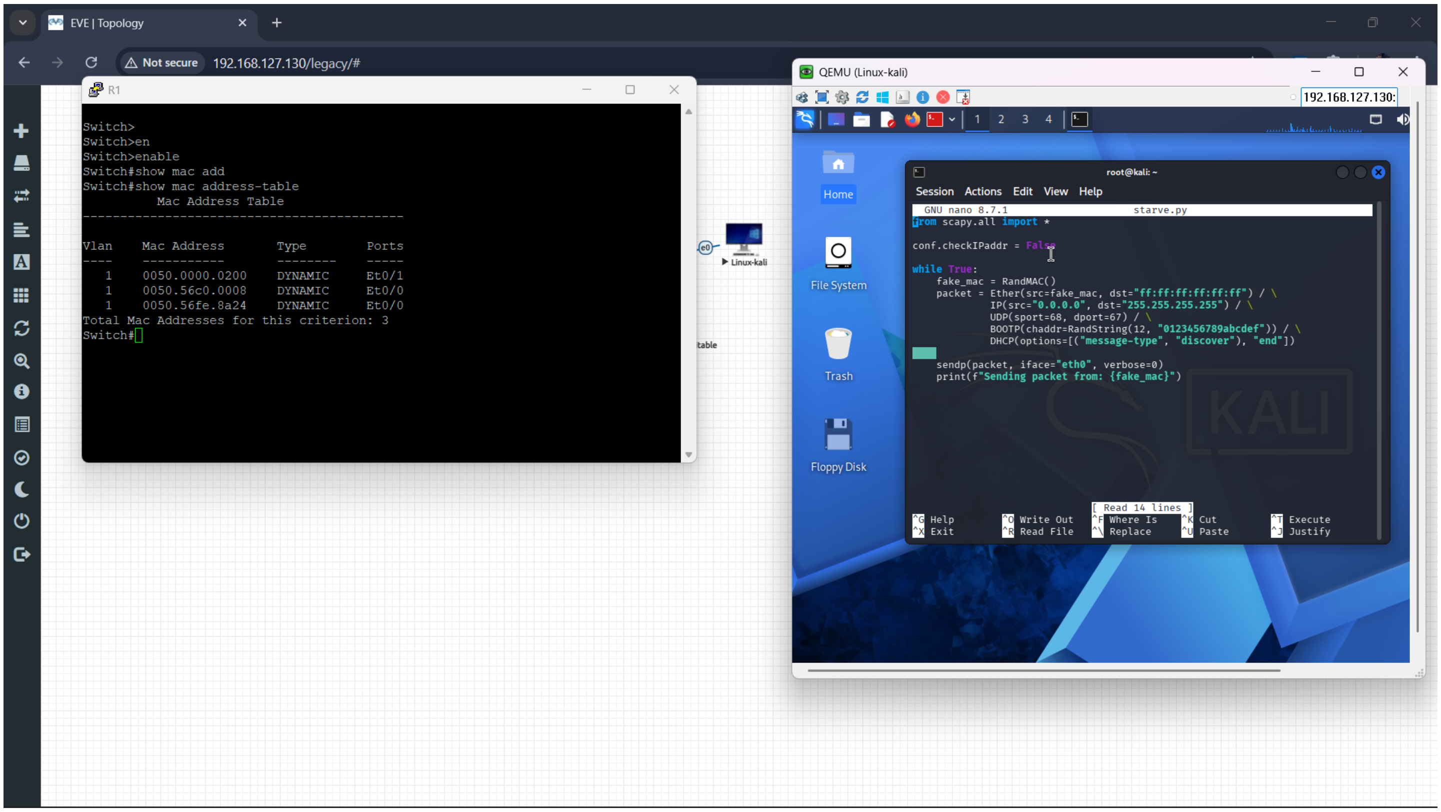
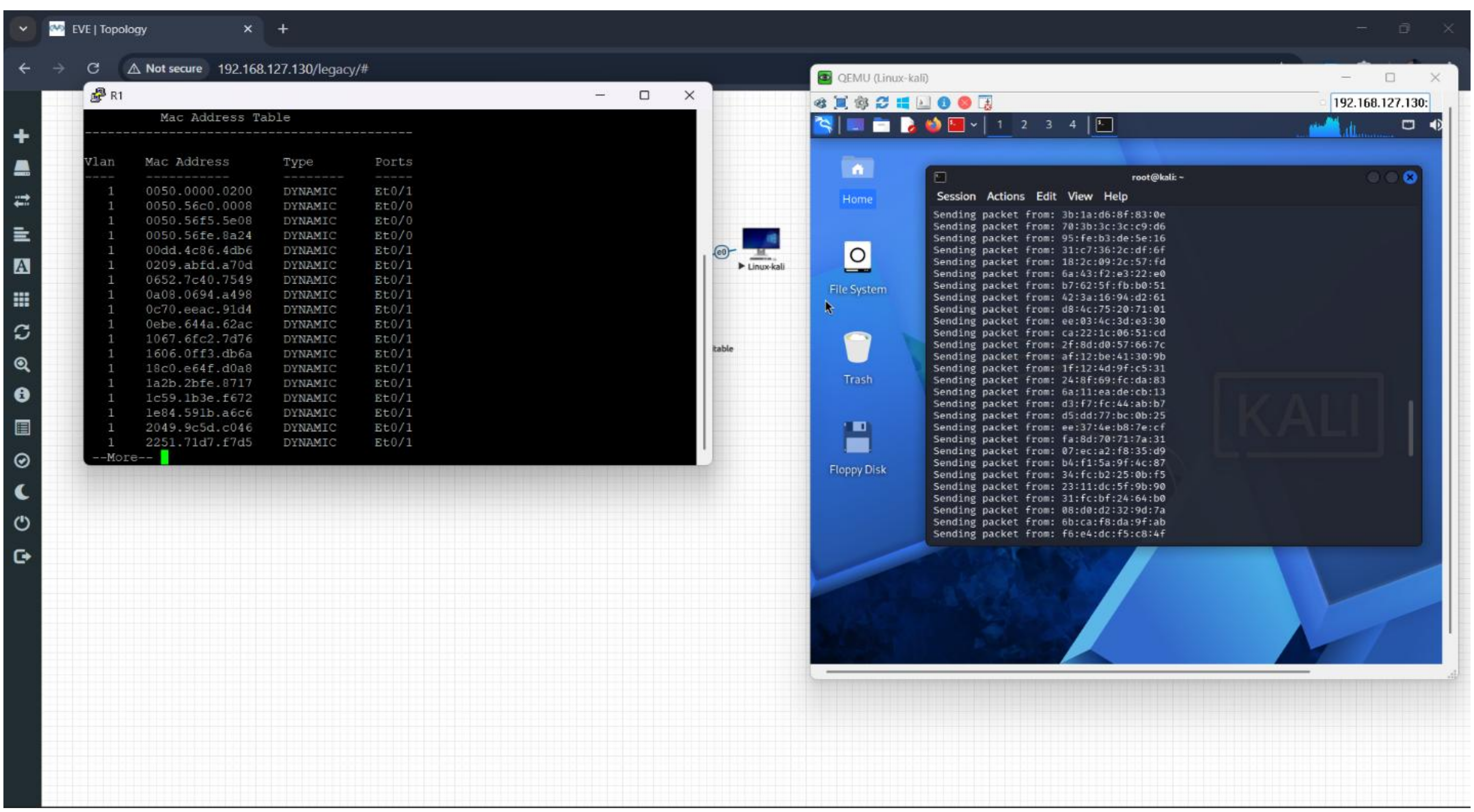
DHCP Starvation Attack

Before Attack:



After Attack :



- **How the Attack Works:**

The attack functions by broadcasting a massive volume of DHCP Discover packets. Each packet is crafted with a unique, randomized Source MAC address and Transaction ID (XID). This tricks the network devices into treating every request as if it originates from a new, unique physical device.

- **DHCP Resource Exhaustion:**

Upon receiving these continuous requests, the DHCP server attempts to allocate an IP address for each one, sending out DHCP Offers. This rapidly consumes the entire IP Pool (available address range). Consequently, the server becomes unable to respond to legitimate clients, leading to a Denial of Service (DoS).

- **MAC Address Table Flooding:**

The switch records the Source MAC of every incoming frame in its CAM Table. By generating random MAC addresses at high speed, the script forces the switch to fill its memory limit. Once the table is full, the switch can no longer perform intelligent forwarding and begins flooding traffic to all ports, essentially acting like a Hub.